NYRA GRAHAM

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EDUCATION

Cornell University | Ithaca, NY
M.Eng., Mechanical Engineering (Robotics), Dec. 2025 | B.S., Mechanical Engineering, May 2025
GPA: 3.62/4.00 | Cum Laude | Dean's List

ENGINEERING EXPERIENCE

Co-Founder & Engineer | Regional Farm Systems | Pennsylvania | *2026*

- Assembled and integrated agricultural drone hardware, including flight controller, onboard computing, camera, power distribution, and communications; supported bench testing and flight demonstrations.
- Architected Demeter, an AI-assisted farm operations platform integrating natural-language workflows, backend tools, and agricultural records.
- Designed an agent-based agricultural marketplace simulation and developed approaches for automated integration and acceptance testing.

Propulsion Component Test Engineering Intern | Relativity Space | Long Beach, CA | *Jun-Aug 2025*

- Independently designed, fabricated, and commissioned a mobile hot-gas valve test rig integrating dual-pressure plumbing, heating, controls, instrumentation, and data acquisition.
- Modified test hardware to reproduce cryogenic valve leakage during thermal shock; developed a visual method to characterize leakage under transient conditions.
- Designed a pneumatic avionics test panel and authored engineering test procedures and reports documenting hardware configurations, anomalies, and results.

Propulsion Engineering Intern | Relativity Space | Long Beach, CA | *Jun-Aug 2024*

- Redesigned an additively manufactured engine test-stand component to reduce bulk and required support material; coordinated load requirements, GD&T, and manufacturability with design and production engineers.
- Redesigned a dual-fluid propulsion injector test rig, integrating fluid plumbing, instrumentation, and imaging to enable simultaneous-flow testing and expand experimental capabilities. Designed removable alignment tooling using tolerance analysis and GD&T

Quality Engineering Intern | Hexcel Corporation | Salt Lake City, UT | *Jun-Aug 2023*

- Designed and implemented production guarding systems and an adjustable test stand for process data collection; developed friction-reduction hardware for fiber manufacturing.

SELECTED PROJECTS

Team Lead / AstroTech Lead / AstroTech Member | Cornell Mars Rover | Ithaca, NY | *2021-2025*

- Led a 65+ member interdisciplinary team through mechanical design reviews, fabrication, prototype iteration, and rover system integration.
- Designed, fabricated, and tested a rover soil-analysis mechanism integrating motor-driven sample chambers, cuvettes, pumps, belt drives, and direction-dependent clutching.
- Integrated and debugged mechanical, electrical, and control systems; supported additive manufacturing, wiring harness fabrication, motor controls, and field testing.

The Weather Machine: Multimodal Haptics | Cornell University | Ithaca, NY | 2025

- Designed the mechanical assembly and prototype enclosure for a weather-simulation device integrating thermal modules, fans, and vibration actuators; performed electrical integration and developed Arduino control software.

TECHNICAL SKILLS

Mechanical Design & Manufacturing: SolidWorks, NX, Inventor, Fusion 360; 3D CAD, ANSYS, GD&T, tolerance analysis, engineering drawings, BOMs, FDM/PBF additive manufacturing, machining, prototyping
Hardware Integration & Validation: Mechanical assemblies, electromechanical mechanisms, test fixtures, instrumentation, DAQ/Dewesoft, DMM, oscilloscope, electrical schematics, soldering
Software & Robotics: Agentic Systems, Python, MATLAB, ROS 2, OpenCV, Open3D, Linux, Git, Raspberry Pi, Arduino